15D 1 CARBOXYLIC ACID DERIVATIVES: ACYL CHLORIDES

SPECIFICATION REFERENCE

15.13 PART 15.14(i) 15.14(ii) 15.14(iii) 15.14(iv)

LEARNING OBJECTIVES

- Understand the nomenclature of acyl chlorides and be able to draw their structural, displayed and skeletal formulae.
- Understand the reactions of acyl chlorides with:
  (i) water
  (ii) alcohols
  (iii) concentrated ammonia
  (iv) amines.

WHAT ARE ACYL CHLORIDES?

These used to be known as acid chlorides, and have the general formula RCOCl. They are derivatives of carboxylic acids, and can be imagined as the result of replacing the OH group in COOH with Cl. Although they contain a carbonyl group (like carboxylic acids, aldehydes and ketones), they are not classed as carbonyl compounds. As with carboxylic acids, having two functional groups sharing the same carbon atom means that the properties are different from both carbonyl compounds and halogenoalkanes.

The commonest example is ethanoyl chloride, CH_3COCl, which is still called by its old name acetyl chloride. You may come across others, such as:

propanoyl chloride CH_3CH_2COCl

and butanoyl chloride $CH_3CH_2CH_2COCl$

DISPLAYED AND SKELETAL FORMULAE

Table A shows the displayed and skeletal formulae of some acyl chlorides.

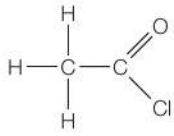
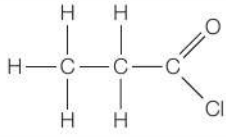
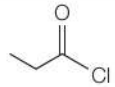
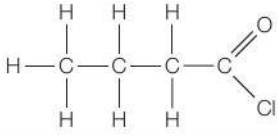
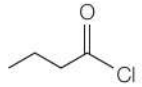
NAME	DISPLAYED FORMULA	SKELETAL FORMULA
ethanoyl chloride		
propanoyl chloride		
butanoyl chloride		

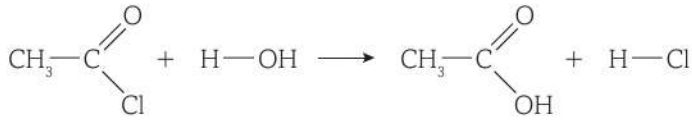
table A

REACTIONS OF ACYL CHLORIDES

The carbon atom in RCOCl is joined to two electronegative atoms, and so is electron-deficient (δ+). You can therefore predict that it will be readily attacked by nucleophiles, such as molecules containing O or N atoms. You need to know about four reactions of acyl chlorides, although no mechanisms are needed for any of them.

REACTION WITH WATER

Acyl chlorides react vigorously (very actively) with cold water, forming a carboxylic acid and releasing hydrogen chloride gas, which appears as misty fumes (**fig A**). The equation for the reaction of ethanoyl chloride with water is as follows:

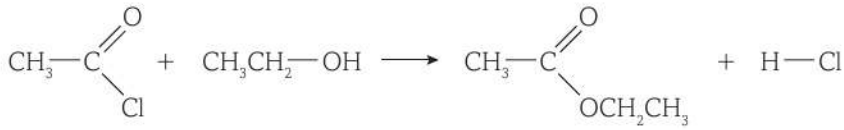




▲ **fig A** Adding ethanoyl chloride to water produces fumes of hydrogen chloride.

REACTION WITH ALCOHOLS

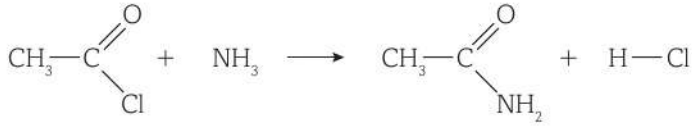
Acyl chlorides react readily with ethanol to form an ester and hydrogen chloride gas. The equation for the reaction of ethanoyl chloride with ethanol is as follows:



Notice the similarity between these two reactions. In both cases, the H of the second reactant combines with Cl to form HCl, and the other part of the reactant becomes joined to the carbonyl group.

REACTION WITH CONCENTRATED AMMONIA SOLUTION

Similar to alcohols, acyl chlorides react readily with concentrated ammonia solution. You can now predict the reaction that occurs: the H of the second reactant combines with Cl to form HCl, and the other part of the reactant becomes joined to the carbonyl group. The NH_2 group joined to the carbonyl group produces a different functional group – amide (not to be confused with amine). The equation for the reaction of ethanoyl chloride with ammonia is as follows:



Unlike with water and alcohols, a further reaction occurs. The reactant is a base and the product is an acidic gas, so these react together to form ammonium chloride:

$NH_3 + HCl \rightarrow NH_4Cl$

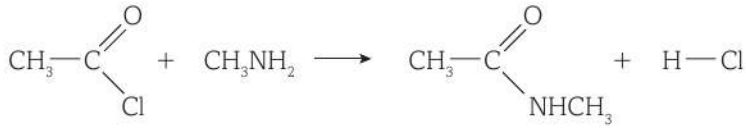
You could write a single equation that combines both of these reactions:

$CH_3COCl + 2NH_3 \rightarrow CH_3CONH_2 + NH_4Cl$

You will learn more about amides later in this book.

REACTION WITH AMINES

This reaction is similar to the previous one. A primary amine can be shown as RNH_2, so you can again predict the reaction that occurs: the H of the second reactant combines with Cl to form HCl, and the other part of the reactant (RNH) becomes joined to the carbonyl group. The equation for the reaction of ethanoyl chloride with the simplest primary amine, methylamine, is as follows:



The product of this reaction is a substituted amide, often called an N-substituted amide. The capital N emphasises that there is an alkyl group attached to N, rather than the usual situation in which it is attached to another C atom. You can see from the product name (N-methylethanamide) that N- is used as a locant, in the same way as in names such as 2-methylbutane – it shows where the methyl group is attached.

Secondary amines (represented as R_2NH) react in the same way, although the product will contain two substituted alkyl groups. A sample equation is:

$CH_3COCl + (CH_3)_2NH \rightarrow CH_3CON(CH_3)_2 + HCl$

The organic product is N,N-dimethylethanamide.

LEARNING TIP

Be sure not to confuse amines (RNH_2) with amides ($RCONH_2$).

You might be able to predict why this type of reaction does not occur with a tertiary amine (R_3N). With three alkyl groups, there is now no H atom to react with Cl to form hydrogen chloride.

CHECKPOINT

SKILLS REASONING

1. Write the names of the organic product formed in the four similar reactions of propanoyl chloride with water, methanol, ammonia and methylamine.
2. Write an equation for butanoyl chloride reacting with:
   (a) propan-1-ol
   (b) ethylamine.

15D 2 CARBOXYLIC ACID DERIVATIVES: ESTERS

SPECIFICATION REFERENCE
15.13 PART 15.15

LEARNING OBJECTIVES

- Understand the nomenclature of esters and be able to draw their structural, displayed and skeletal formulae.
- Understand the hydrolysis reactions of esters, in acidic and alkaline solution.

INTRODUCTION

In previous sections, you have come across esters as the products of reactions of other organic compounds. We will now take a closer look at esters, including their reactions.

NAMING ESTERS

This is straightforward, but you need to remember these points:

- As for carboxylic acids and acyl chlorides (but not amides), an ester name contains two words.
- The first word comes from the alkyl group joined to O.
- The second word comes from the alkyl group joined to C.

Table A shows some examples.

STRUCTURAL FORMULA	NAME
$HCOOCH_3$	methyl methanoate
CH_3COOCH_3	methyl ethanoate
$HCOOCH_2CH_3$	ethyl methanoate
$CH_3COOCH_2CH_3$	ethyl ethanoate
$CH_3COOCH_2CH_2CH_3$	propyl ethanoate

table A

DISPLAYED AND SKELETAL FORMULAE

Table B shows the displayed and skeletal formulae of some esters.

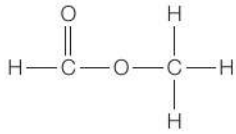
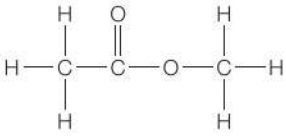
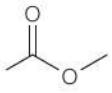
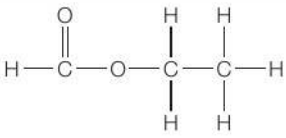
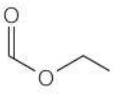
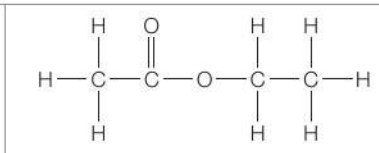
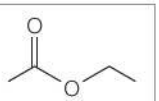
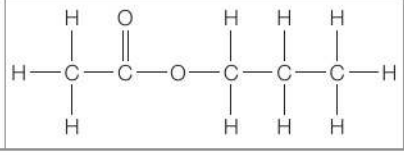
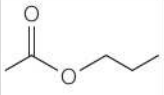
NAME	DISPLAYED FORMULA	SKELETAL FORMULA
methyl methanoate		
methyl ethanoate		
ethyl methanoate		
ethyl ethanoate		
propyl ethanoate		

table B

PHYSICAL PROPERTIES

Esters are colourless liquids with relatively low melting and boiling temperatures, and are insoluble in water. All of the hydrogen atoms in their molecules are attached to carbon atoms, so hydrogen bonding is not possible.

The generally pleasant smells of esters are largely responsible for the familiar odours of flowers and fruits. Examples include:

pentyl ethanoate	pears
3-methylbutyl ethanoate	bananas
methyl butanoate	apples

Esters are present in perfumes, food flavourings, solvents, anaesthetics and biofuels.

DID YOU KNOW?

Benzyl ethanoate is a component of food flavourings that give the taste of apples and pears.



▲ **fig A** The familiar odours of apples and pears come from the generally pleasant smells of esters.

HYDROLYSIS OF ESTERS

The only reactions of esters that you need to know about are two slightly different types of **hydrolysis**. The term 'hydrolysis' means

breaking down (*lysis*) using water (*hydro*). It can be considered as the reverse of esterification. You may remember equations such as this from a previous section:

$$CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOCH_2CH_3 + H_2O$$

This shows the formation of ethyl ethanoate from the corresponding carboxylic acid and alcohol. The reversible arrow indicates that a mixture of ethyl ethanoate and water reacts together to form ethanoic acid and ethanol. This reverse reaction could be described as the hydrolysis of ethyl ethanoate.

HYDROLYSIS IN ACIDIC SOLUTION

We partly covered this type of hydrolysis in the previous paragraph. For a given ester, warming it with water will cause hydrolysis. As with esterification, the reaction is slow. A catalyst such as sulfuric acid will speed up hydrolysis, but will not affect the position of equilibrium, so the reaction will not go to completion:

$$CH_3COOCH_2CH_3 + H_2O \rightleftharpoons CH_3COOH + CH_3CH_2OH$$

HYDROLYSIS IN ALKALINE SOLUTION

The disadvantage of using an alkali such as aqueous sodium hydroxide instead of sulfuric acid is that the reaction produces a carboxylate salt instead of a carboxylic acid. However, the advantage easily outweighs this disadvantage – the reaction goes to completion instead of reaching an equilibrium. The equation for methyl propanoate is:

$$CH_3CH_2COOCH_3 + NaOH \rightarrow CH_3CH_2COO^- + Na^+ + CH_3OH$$

Converting the carboxylate salt into a carboxylic acid only needs a dilute acid to be added:

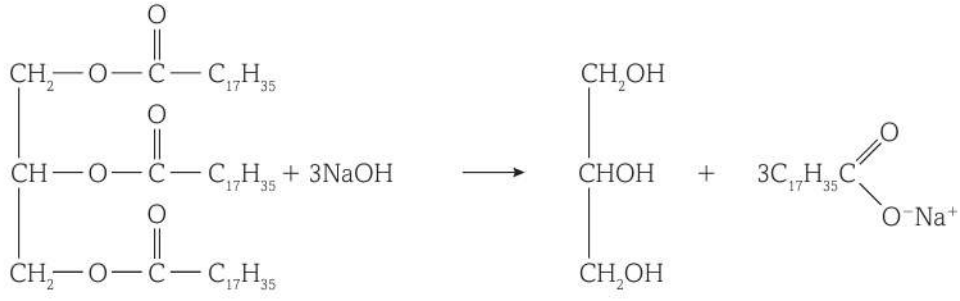
$$CH_3CH_2COO^- + H^+ \rightarrow CH_3CH_2COOH$$

The final organic products are methanol and propanoic acid.

SAPONIFICATION

The Latin word for soap is *sapo*, so saponification means 'soap-making'. This is a particular example of the alkaline hydrolysis of esters. Soaps and detergents have similar uses, but the main difference is that detergents use organic compounds obtained from crude oil, while soaps use organic compounds obtained from vegetable (and, originally, animal) sources.

Vegetable oils contain large quantities of triglycerides. These are triesters, each of which consists of a large ester molecule that can be hydrolysed to one alcohol and three carboxylic acid molecules. The hydrolysis of a typical triglyceride is shown in **fig B**.



a triglyceride: an ester of long-chain carboxylic acids and a triol

propane-1,2,3-triol (glycerol)

sodium octadecanoate (sodium stearate)

▲ **fig B** The hydrolysis of a typical triglyceride is an example of saponification.

LEARNING TIP

Acidic hydrolysis of an ester is a reversible reaction, but alkaline hydrolysis reactions go to completion.

In this example, hydrolysis produces an alcohol with three hydroxyl groups (glycerol). Glycerol has many uses, including as a sweetener in foods and toothpastes, and in skin care products. It is also used as a component of the liquid in electronic cigarettes.

The other product of this reaction, commonly known as sodium stearate, is a very common ingredient of most soaps.

CHECKPOINT

1. What are the names of these esters?
   (a) $CH_3CH_2COOCH_2CH_2CH_3$
   (b) $(CH_3)_2CHCOOCH_3$
2. Write equations for the hydrolysis of propyl butanoate in:
   (a) acidic conditions
   (b) alkaline conditions.

SUBJECT VOCABULARY

hydrolysis the breaking of a compound by water into two compounds

15D 3 CARBOXYLIC ACID DERIVATIVES: POLYESTERS

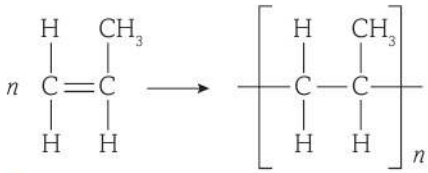
SPECIFICATION REFERENCE 15.16

LEARNING OBJECTIVES

- Understand how polyesters, such as terylene, are formed by condensation polymerisation reactions.

ADDITION POLYMERISATION

Examples of addition polymerisation were covered in **Topic 5 (Book 1: IAS)**, so may be useful for reminding you of this type of polymerisation. The monomer propene can combine with many thousands of other propene molecules to form a very long chain. This process can be represented as:



▲ **fig A** This equation represents the polymerisation of propene.

One thing to notice is that all of the atoms in the monomer molecules end up in the structure of the polymer.

CONDENSATION POLYMERISATION

There are two main differences between addition polymerisation and **condensation polymerisation**. In condensation polymerisation:

- each time two monomer molecules join together, another small molecule is formed
- usually two different monomers react together.

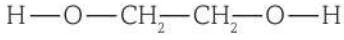
In many examples of condensation polymerisation, the small molecule formed is water, which is the origin of the term 'condensation'. In other examples, the small molecule formed is hydrogen chloride, but the term 'condensation' is still used to describe those examples.

One type of polymer formed in this way is a polyamide, which we will meet in **Topic 19**.

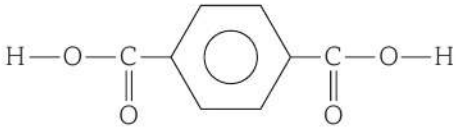
POLYESTERS

The example we will use to introduce condensation polymerisation is the formation of polyesters. You will remember that esters are the products of the reaction between alcohols and carboxylic acids. However, once one molecule of an alcohol has reacted with one molecule of a carboxylic acid, the reaction is complete. The only organic product is a slightly larger ester molecule.

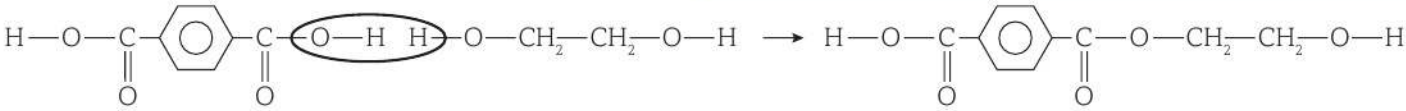
For a reaction to produce a polymer, we need two monomers, each with two reactive groups (one at each end). The alcohol usually involved in this type of reaction, more accurately called a 'diol', has two hydroxyl groups, one at each end:



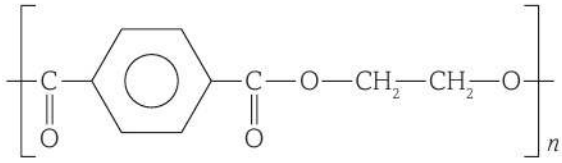
One of the most common carboxylic acids used (more accurately a dicarboxylic acid) has the structure below. The common name for this compound is terephthalic acid.



Consider how these molecules react together. The OH of the carboxylic acid and the H of the alcohol react together to form the small molecule, water, and the two molecules are linked together by an ester group. This equation shows what happens during the formation of a typical polyester.



This larger molecule that has formed still has reactive groups at both ends. This means it can continue reacting with other molecules in the same way as above, until a very long polymer chain has formed. The structure of the polymer is:



Because of the original name of the acid (terephthalic acid), the polymer was originally known as Terylene.

Polyesters have a very wide range of uses, including soft drink bottles, food packaging, many types of clothing and duvet fillings.

It is also possible to use a dioyl chloride instead of a dicarboxylic acid, although this is not used as a monomer in industry. The reaction occurs in the same way – the only difference is that the small molecule formed is hydrogen chloride, not water.

EXAM HINT

You may like to think about why dicarboxylic acids rather than dioyl chlorides are used in industry. Hint: look back at **Topic 4 (Book 1: IAS)** and think about hazards and risks.

LEARNING TIP

When showing how a polyester forms, remember that, in the formation of the small molecule, the H comes from the diol and the OH from the dicarboxylic acid (or the Cl from the dioyl chloride).

CHECKPOINT

1. Why is it not possible to make a polymer by reacting together molecules of $HOOCCH_2COOH$ and CH_3OH?
2. Draw the repeat unit of the polymer formed between molecules of HOOCCOOH and $CH_3CH(OH)CH(OH)CH_3$.

SUBJECT VOCABULARY

condensation polymerisation the formation of a polymer, usually by the reaction of two different monomers, when another small molecule is also formed